

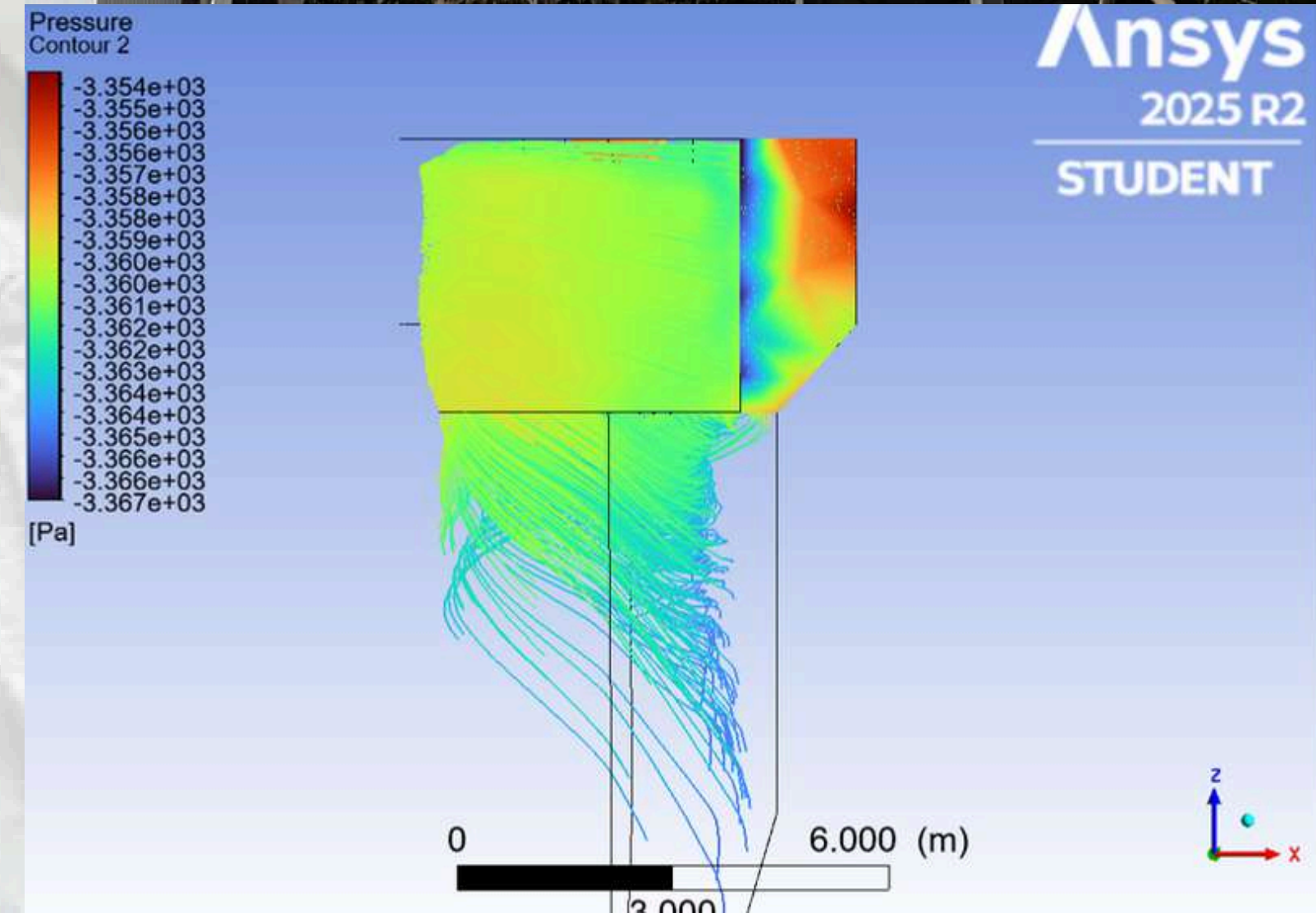
Mechanical Engineering Portfolio

Zaki Fahrizal

Linkedin : Zaki Fahrizal

Fluid Dynamics Analysis on Suspension Preheater (Top Cyclone) for Cement Production

Process Engineering Intern | Baturaja Portland Cement, Indonesia
Dec 2025 - Present



What is Cyclone?

A cyclone is a tool used in the cement industry to separate gases and solids from combustion products in a rotary kiln. Centrifugation is expected to separate the solids from the gas, preventing environmental pollution and resulting in a more efficient process because all solids are returned to the kiln without the need for filtration. The gas inlet contains both gas and solids (combustion dust and kiln feed). 100% efficiency is achieved when all solids exit through the bottom outlet.

Problem:

The pressure drop now requires significant fan power to draw air out of the cyclone. Installing a Flow Deflector and modifying the Vortex Finder design is expected to reduce this in three ways:

- 1. Increase Separation Efficiency
- 2. Reduce Heat Loss (Turbulence)
- 3. Reduce Pressure Drop at the Inlet and Upper Outlet

Solution:

The study was conducted on several cement factory evaluations in Indonesia. To reduce the effects of turbulence and straighten air flow, several design modifications to the flow deflector and vortex finder on the upper preheater were concluded, as follows:

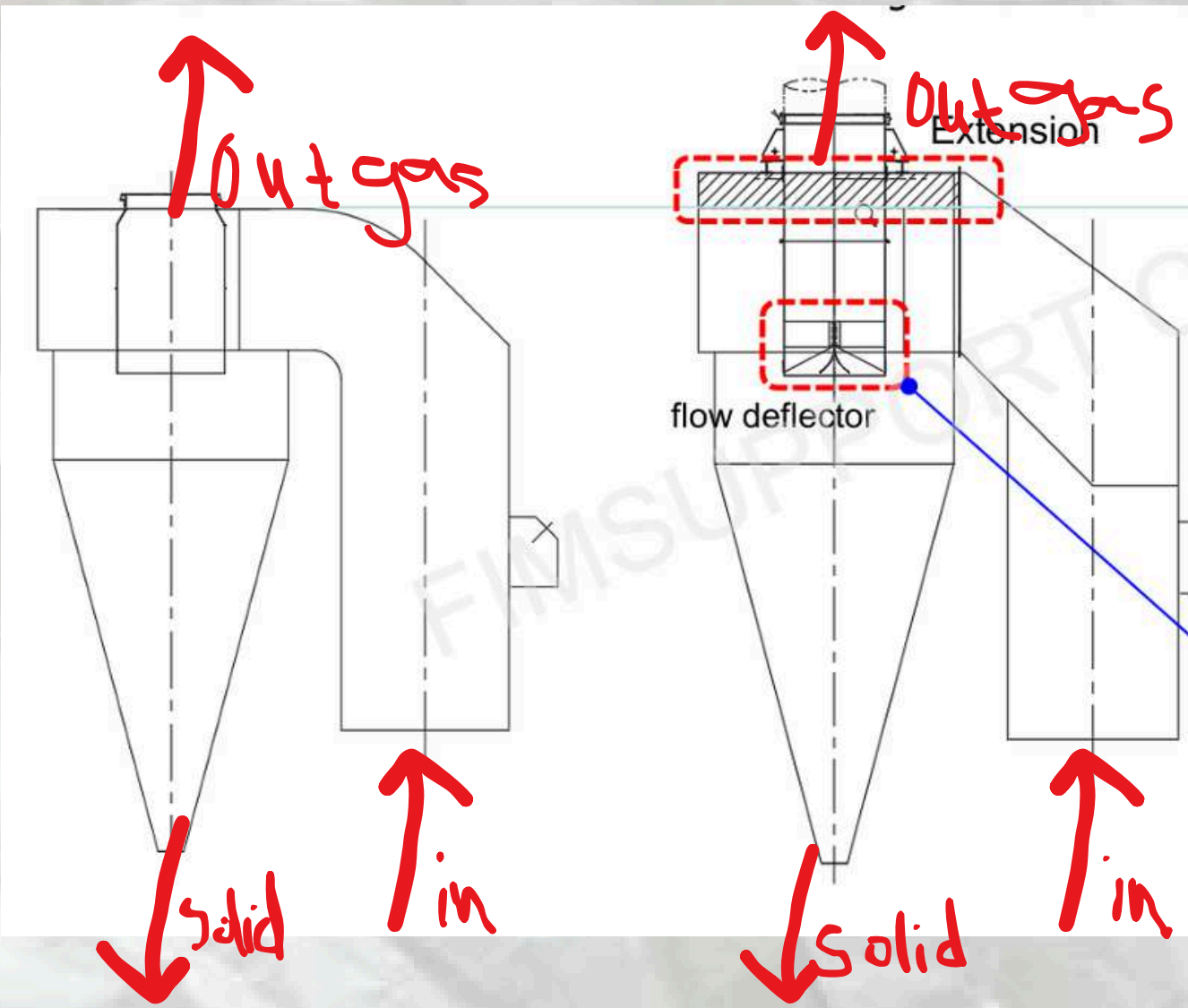
- 1. Original design
- 2. 1/4-section flow deflector (along the vortex finder)
- 3. 1/3-section flow deflector (1 meter)
- 4. 1-meter extension of the vortex finder
- 5. 1-meter extension of the vortex finder
- 6. 1/3-section flow deflector (curved) (1 meter)

Method & Evaluation :

Designs are compared based on three aspects (Separation Efficiency, Heat Loss, and Pressure Drop). These three aspects affect the primary function of a cyclone, and losses from these three aspects will be calculated to determine the most economical design.

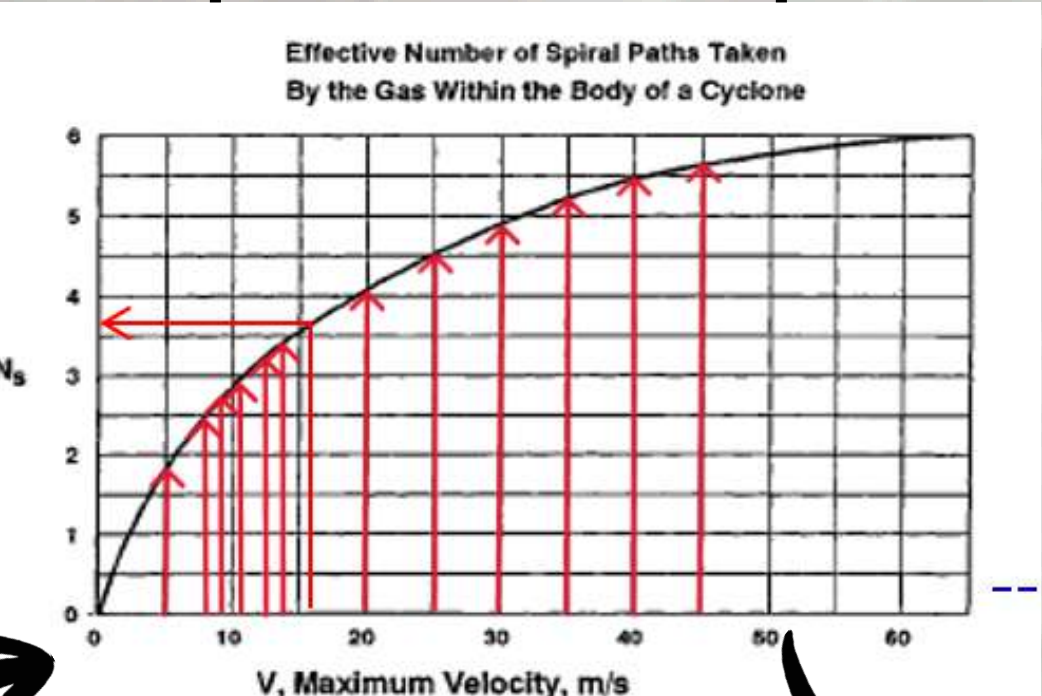
Current Measured Pressure and Temperature accross Cyclone

ILC									
Temp, Press, O2, CO Profile		Kilnline							
		Calc. Outlet	A55	A54	A53	A52	A51	A61	Tower exit
Gas temperature	[°C]	902.00	0.00	902.00	787.00	642.00	338.90	325.81	371.00
Pressure	[mbar]	-19.44	-21.47	-23.50	-38.50	-50.50	-71.89	-72.01	-72.01
dp	[mbar]	12.50	14.53	2.03	15.00	12.00	21.39	21.51	0.06
dp: Design Criteria			<6	<9	<9	<9	9-14	9-14	
Oxygen	[%]	1.28	1.50	1.50	1.79	2.08	2.64	2.65	2.65

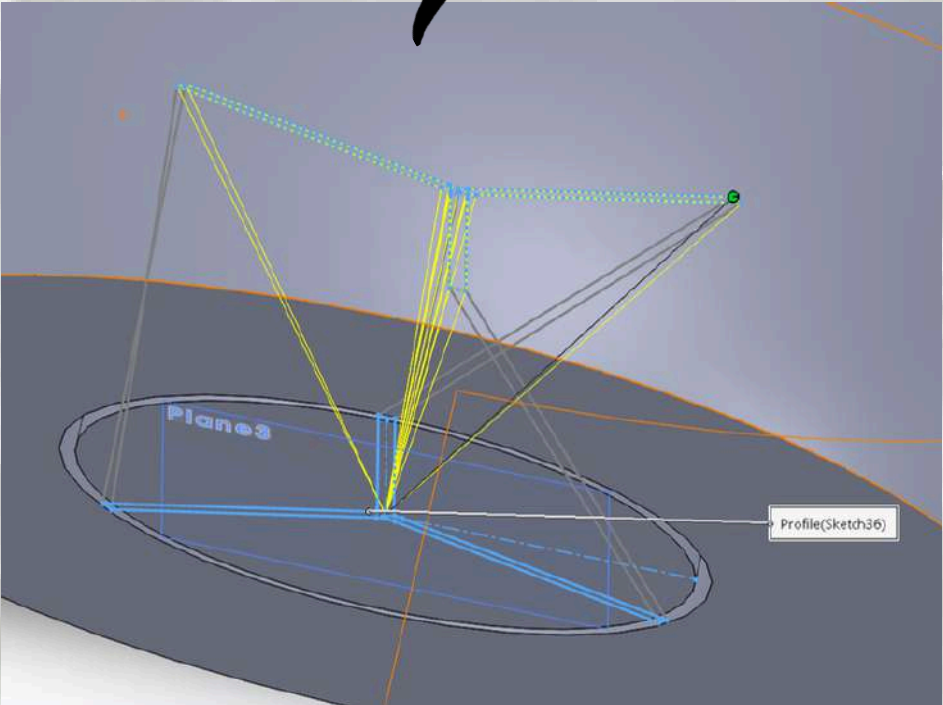
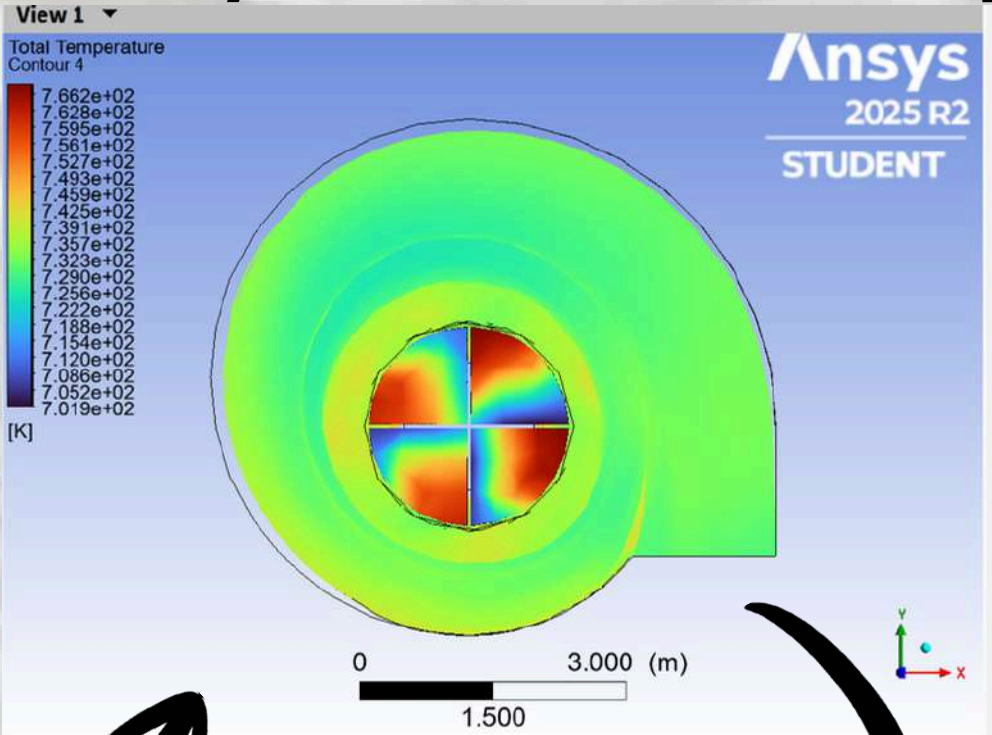


Top Cyclone flowchart

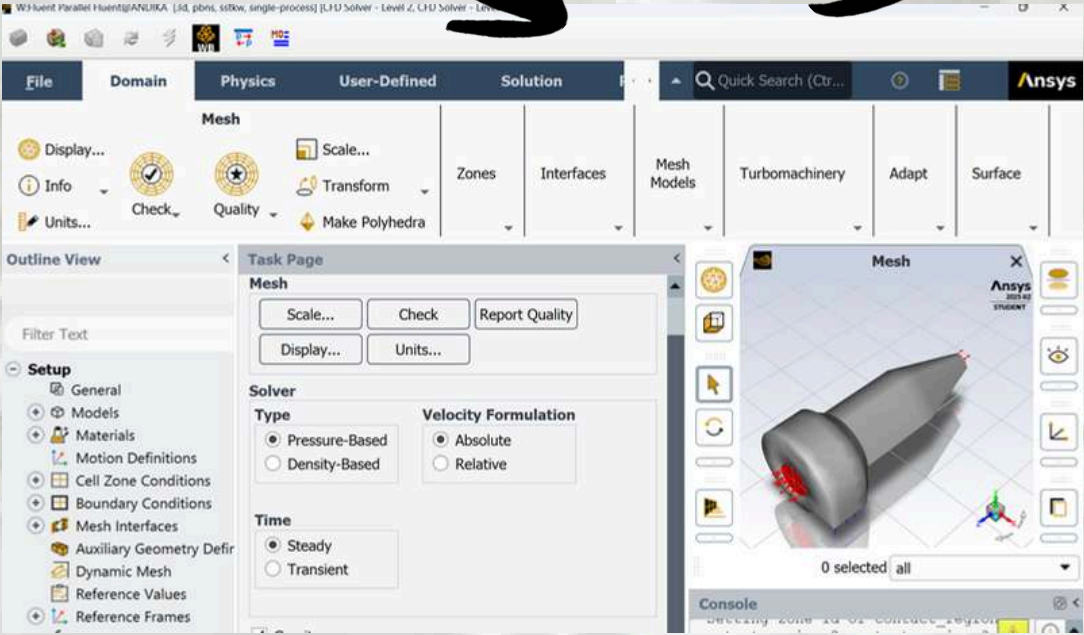
Theoretical Calculation of Separation Efficiency and Experimental Assumption



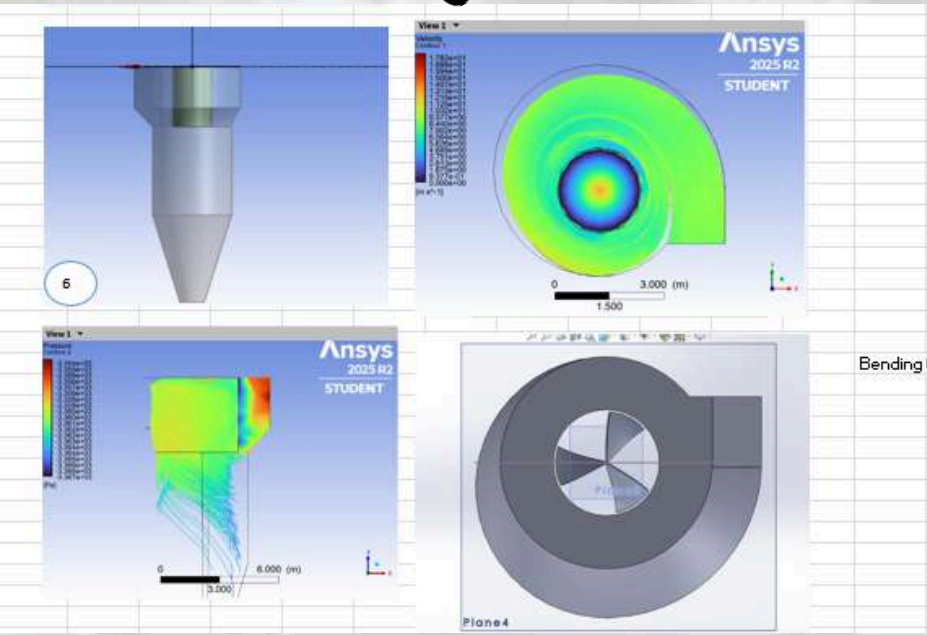
Result + Validation (Efficiency, Heat Loss, Pressure Drop)



CAD Design + Flow Deflector Assembly

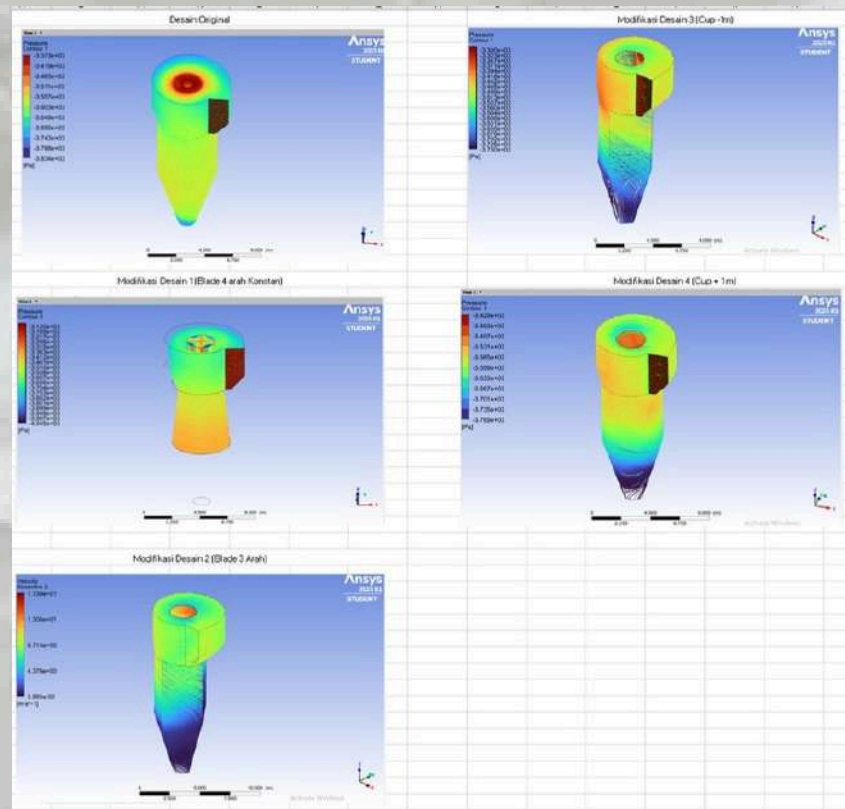


CFD Modelling (ANSYS)

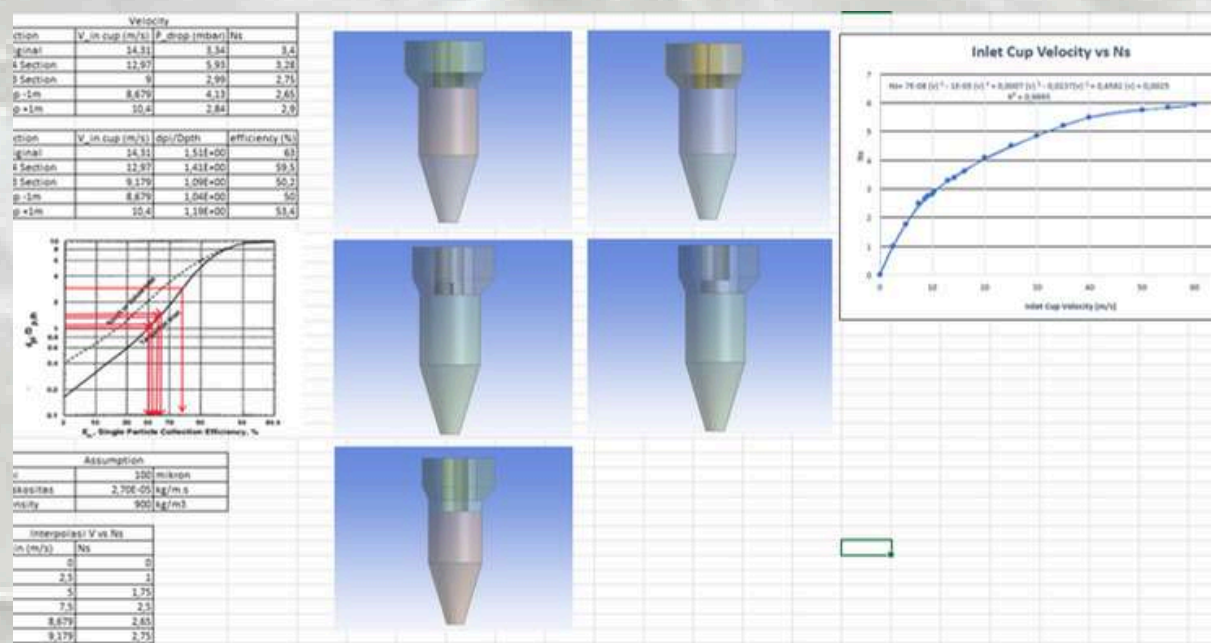


Design Cost Loss Comparison

Result and Optimisation



CFD Result



Design Comparison and Theory Validation



Pricelist				
Parameter	Amount	Unit		
FC Cost	Rp 1.047.609,00	IDR/ton		
FC Heat Value	4500	kcal/kg FC		
Sp. Cost of FC	4,295495743	kcal/IDR		
Listrik	Rp 998,00	IDR/kwh		
Clinker Price	Rp 410.000,00	IDR/ton		
SEEC Kiln System	24	kwh/ton Cl DB		
Transport Loss	Rp 23.952,00	IDR/ton Cl DB		
List of Cyclone Losses / hour				
Desain	Loss (p_drop)	Loss (Heat)	Loss (Separation)	Net Loss
Original	Rp 22.831,16	Rp 94.911	Rp 498.968,06	Rp 616.709,85
1/4 Section	Rp 40.563,41	Rp 265.698	Rp 513.438,85	Rp 819.700,43
1/3 Section	Rp 20.442,71	Rp 139.593	Rp 857.845,98	Rp 1.017.882,15
Cup -1m	Rp 28.233,07	Rp 112.388	Rp 637.273,23	Rp 777.894,02
Cup +1m	Rp 19.415,28	Rp 116.385	Rp 734.001,52	Rp 869.801,57
Curved	Rp 24.143,64	Rp 95.940	Rp 755.236,25	Rp 875.319,95

Cost Calculation for Each Design

Result

The current design demonstrates the best performance in terms of separation efficiency and heat loss. While the pressure drop does decrease with the 1/3 section design modification, the resulting separation efficiency is not as optimal as the original design. Separation losses significantly exceed the optimized pressure drop.

Current Progress /Improvement

The flow deflector geometry will be studied more specifically by surveying more designs to minimize turbulence effects. Dispersion modeling will also be performed to validate the results of the separation efficiency calculations.

What I have Learn

I learned a lot about optimizing design and geometry on a real production tool, and by presenting the results of my studies to senior engineers, I felt I had made a real contribution and developed my engineering thinking.



Thermo-Structural Analysis with Internal Combustion Expansion Effect on Refractory Brick for Rotary Kiln

**Process Engineering Intern | Baturaja Portland Cement, Indonesia
Dec 2025 - Present**

Background, Problem, and Solution



Kiln Inspection to track Crack Point



Refractory Brick Crack

Background

The purpose of installing refractory bricks is to retain the heat from combustion inside the kiln, thereby reducing heat loss. Within the kiln structure, refractory bricks experience several forces triggered by significant temperature differences between the outside and inside. The bricks will expand and eventually crack. These cracks will cause uneven heat transfer. As a result, the outer layer (shell) will receive more heat than usual and expand more rapidly, shortening its lifespan.

Problem

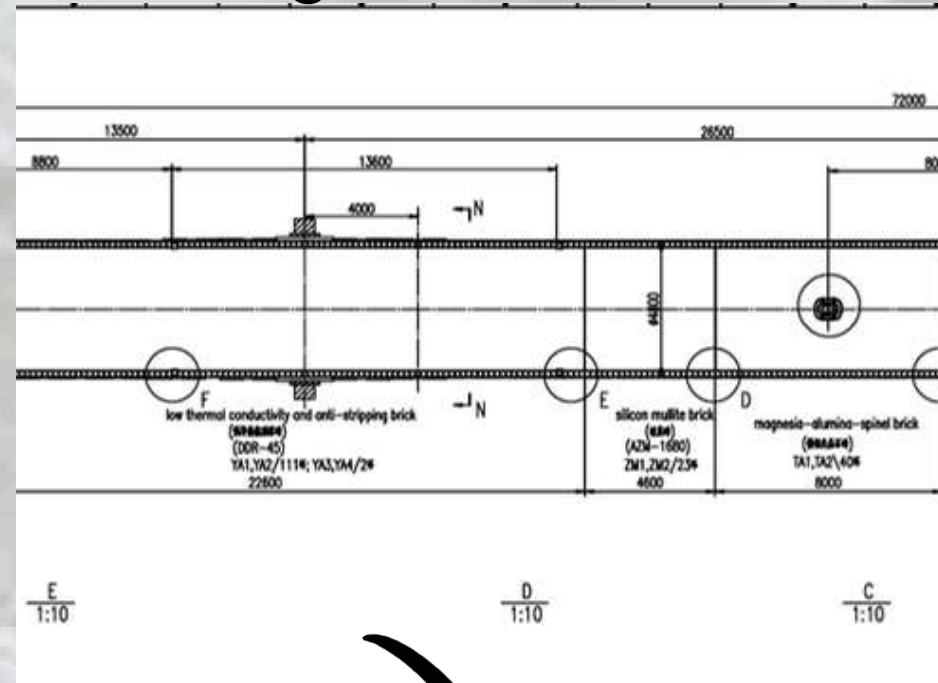
The shell can experience a shorter lifespan due to thinning bricks.

Solution

The position of the shell's temperature distribution and its expansion effects can be tracked, so that the shell can be further optimized (e.g., adding stiffeners or replacing it with a stronger type of steel).

Method

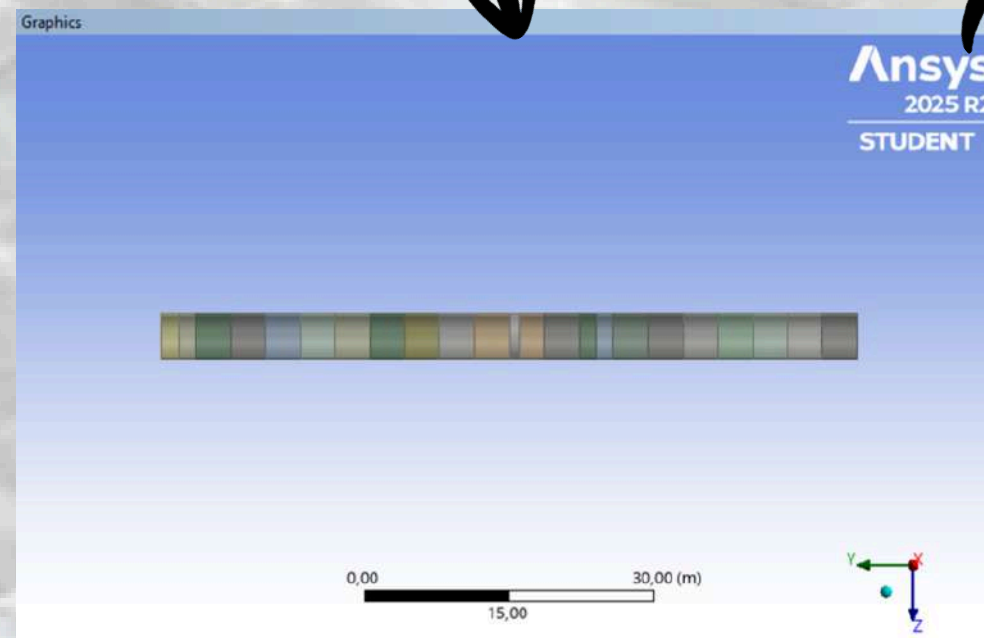
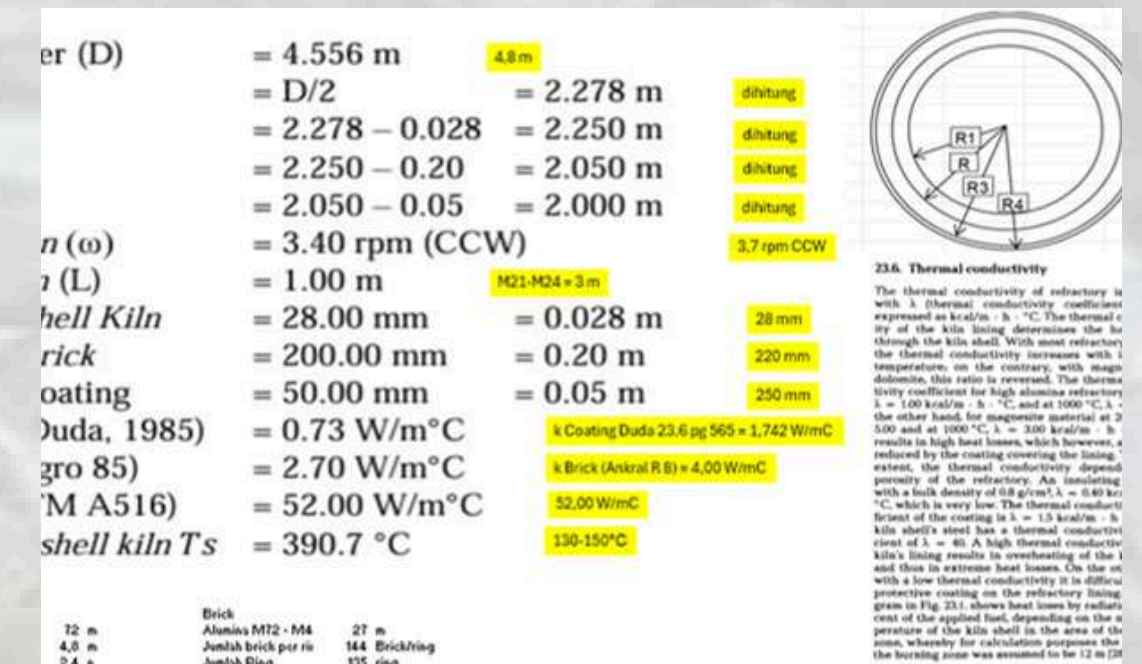
CAD Design Review and Drawing



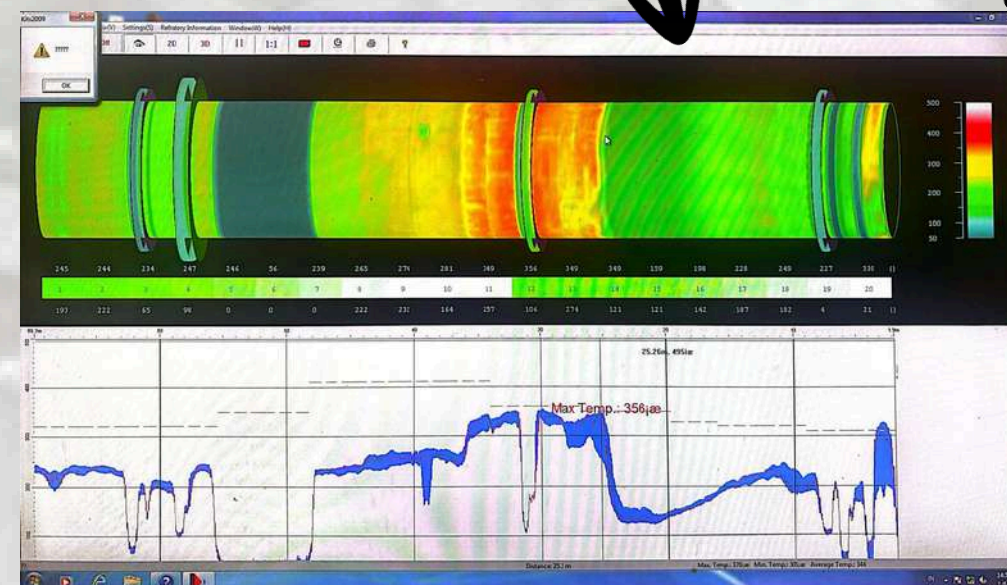
Brick and Shell Material Inserted based on Real Life

Properties of Outline Row 3: ASTM A516					
	A	B	C	D	E
1	Property	Value	Unit		
2	Material Field Variables	Table			
3	Density	7,85	g cm ⁻³		
4	Isotropic Secant Coefficient of Thermal Expansion				
5	Coefficient of Thermal Expansion	1,2E-05	C ⁻¹		
6	Isotropic Elasticity				
7	Derive from	Youn...			
8	Young's Modulus	200	GPa		
9	Poisson's Ratio	0,29			
10	Bulk Modulus	1,5873E+11	Pa		
11	Shear Modulus	7,7519E+10	Pa		
12	Tensile Yield Strength	260	MPa		
13	Tensile Ultimate Strength	485	MPa		
14	Isotropic Thermal Conductivity	52	W m ⁻¹ ...		

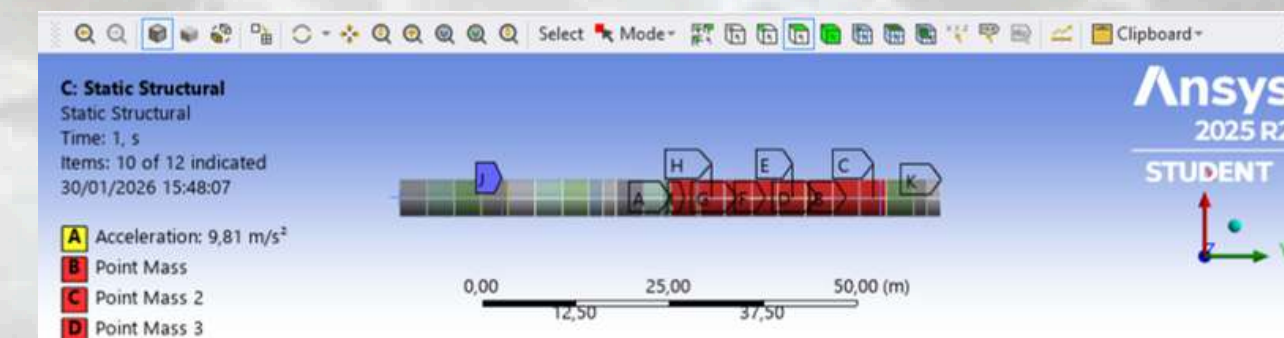
Theoretical Calculation of Heat Flux for Boundary Condition



Kiln Divided into 4 main Zone (Inlet, Burning, Cooling, Outlet)

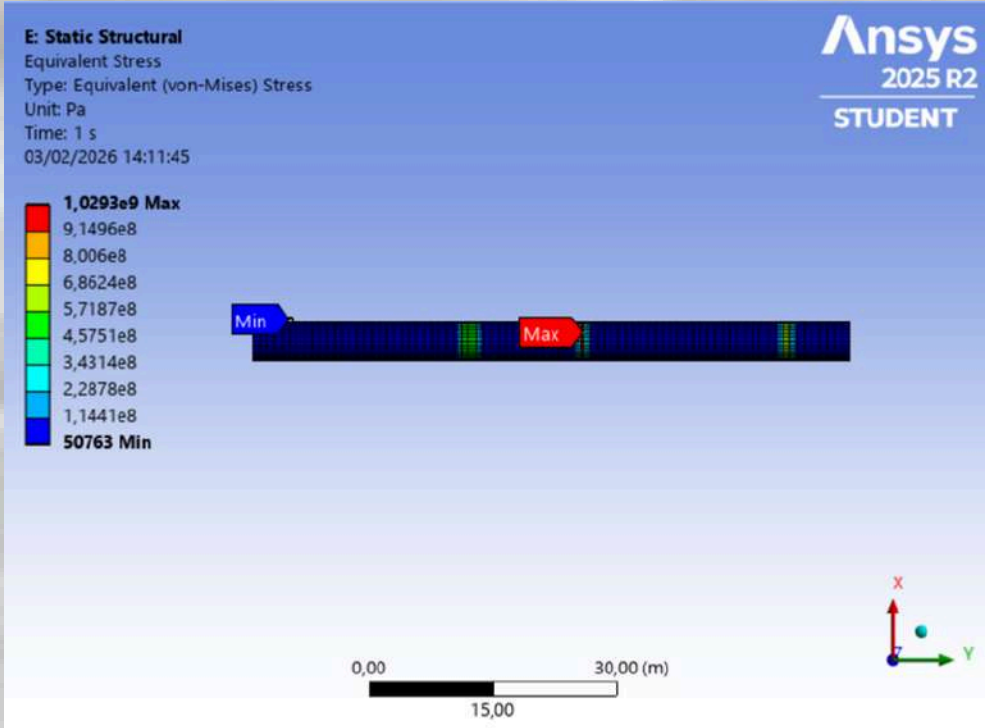


Boundary Condition (Heat Flux, Convection, Coating Thickness) Calculated based on real measurement using Infrared Sensor

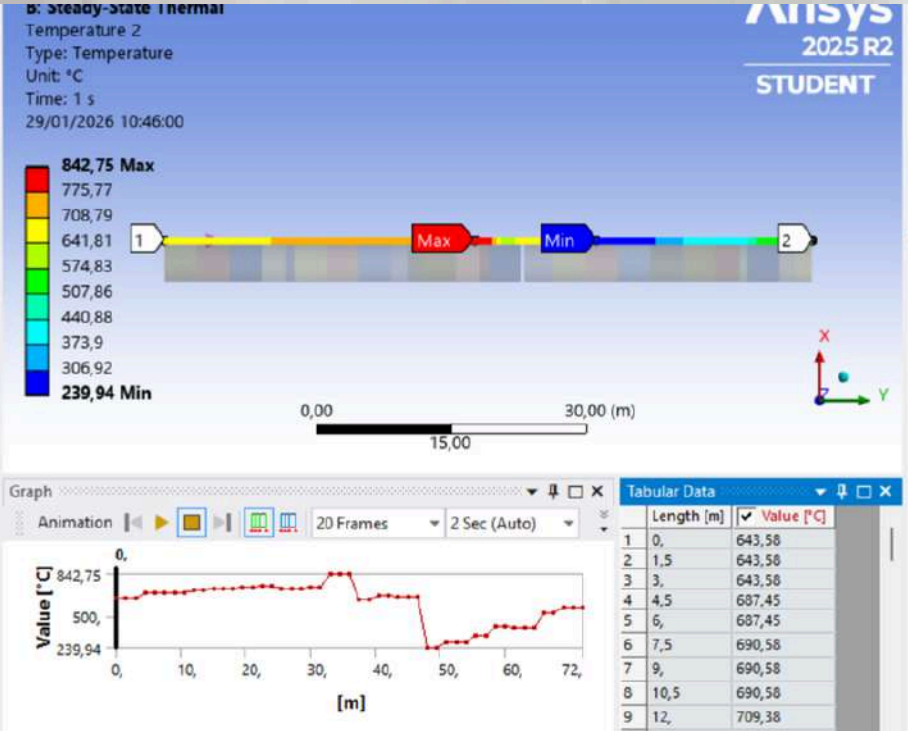


Final Module Result

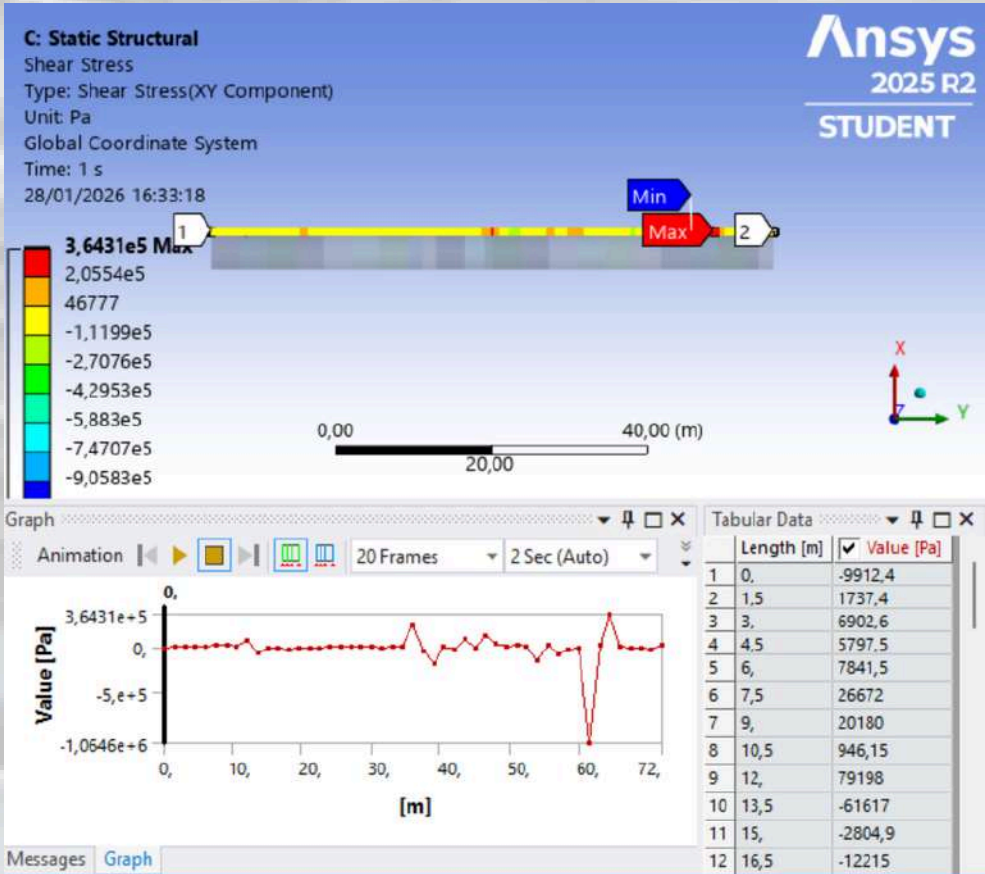
RESULT



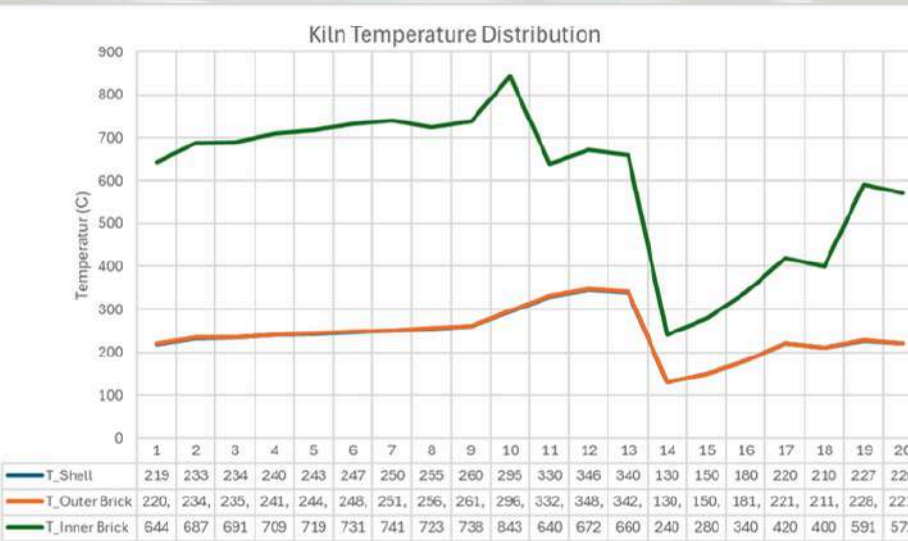
Stress Distibution



Refractory Brick temperature Distribution



Shear Stress Distribution

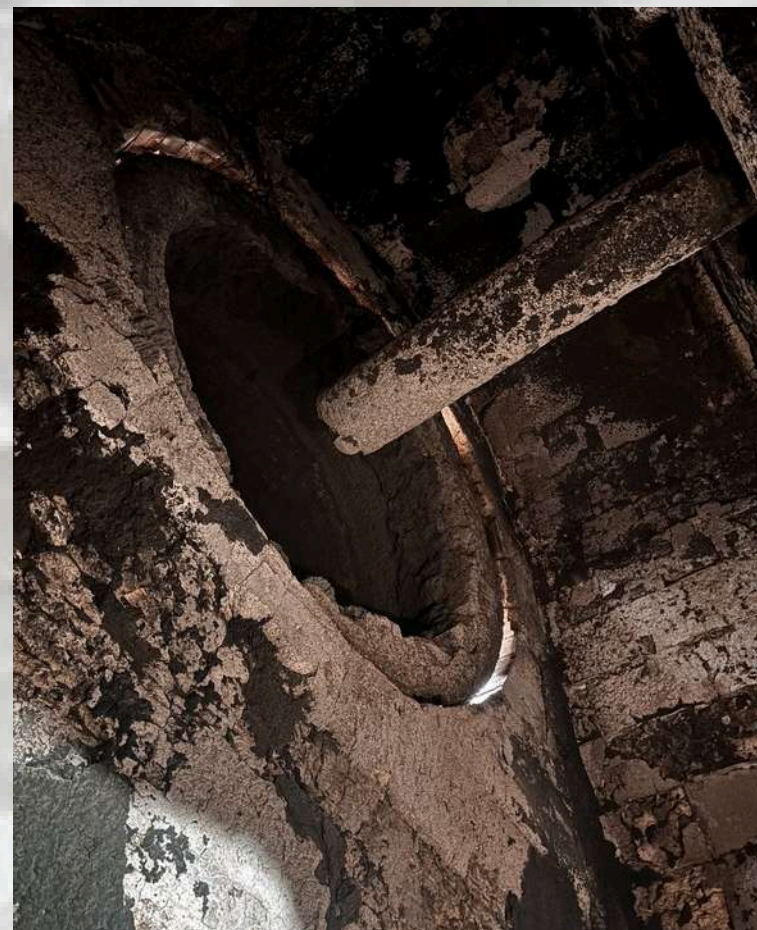


Result:

1. The temperature distribution was successful on the track, with the highest temperature occurring at meter 36. The temperature obtained in the outer refractory brick layer was around 800–900 degrees Celsius.
2. The maximum stress was identified at 35–39 MPa. The stress on the support was ignored because it was considered fixed, and only the effects of expansion were considered. It was concluded that the design was still safe based on the shear stress, maximum von-Mises stress, bending stress less than the compressive stress, and ultimate tensile strength (around 400 MPa).
3. The coating thickness resulting from the melted clinker can be estimated through conduction and heat transfer calculations. This thickness can be useful during inspections; if the thickness is less than the specified calculation, it can be patched to maintain shell temperature stability.



Refractory Brick Inspection



Burner



Coaring on Burning Zone



Inspection)

Conclusion

How this Evaluation can be beneficial to company?

This calculation serves as a basis for determining a safe coating level in the kiln blasting zone, where bricks are susceptible to cracking due to the high gas temperature (1450 degrees Celsius). Therefore, the expansion effect can be optimized by selecting the right brick, preventing spontaneous expansion and ensuring the shell supports the kiln properly.

Shell material selection can also be based on the external temperature distribution, allowing for the estimation of cracking points in the shell and subsequent reinforcement with stiffeners.

What I Learn?

I learned how the expansion of high-temperature gases can affect the mechanical performance of a rotary kiln. I hope that measuring the refractory brick and shell expansion will enable companies to optimize the temperature control of the inner workings of rotary kilns, ensuring optimal firing temperatures.

sUNSWim Robotics @ Mechanical Team (Sep 2025 – Present)

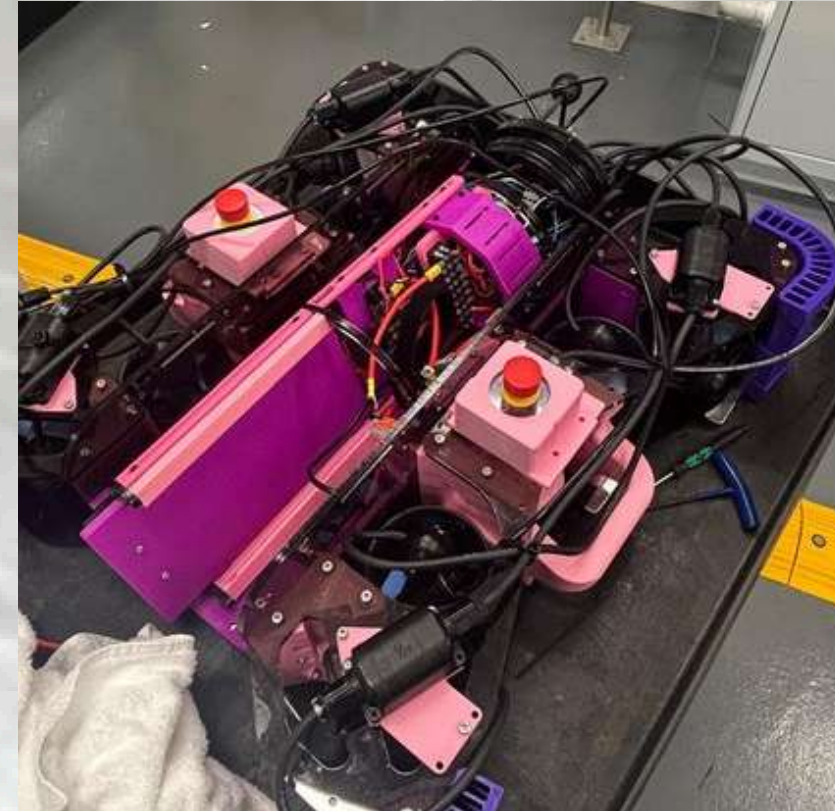
Involved as Mechanical Team at sUNSWim, UNSW Underwater Robotics Team. Here, me and my teammates design and operate an ROV (Remotely Operated Vehicle) using 3-d Printed Materials, and validates the design through pool testing. This Design is Joining Yearly SAUVC, and now gearing up for SAUVC (Singaporean Autonomous Underwater vehicle Championship) 2026. I enrolled on this program on VIP ENGG4600.



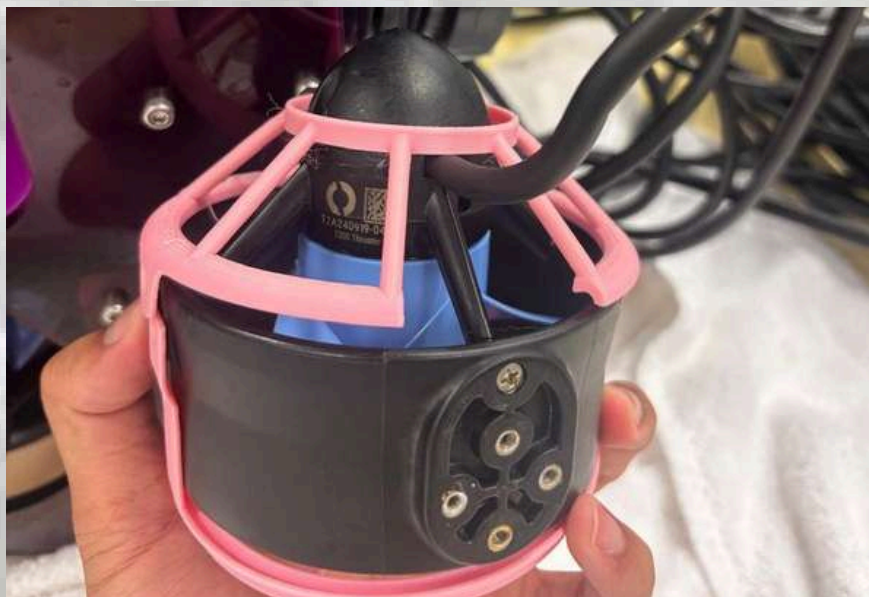
Hands-on Progress



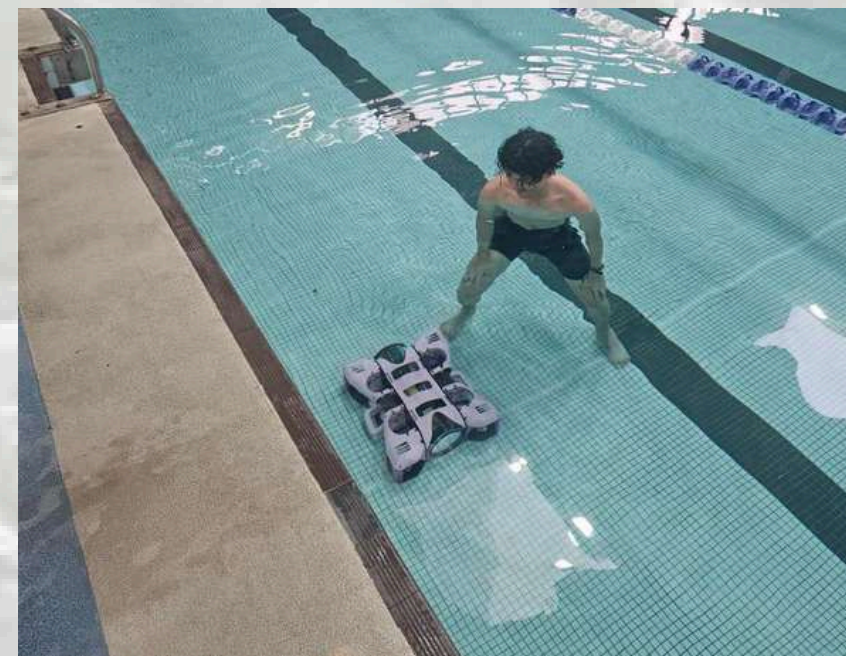
3D Printed Designs



Robot Cable Management, Component Assembly



Thruster Guard Assembly and Propeller Testing



Buoyancy and Drag Pool Test

The design will be 3D printed and tested under swimming pool conditions (in accordance with the original Competition conditions) to test strength, stability and the effect of additional equipment on the mechanical performance of the vessel (buoyancy, drag and mounting mechanisms).



What I have Learn

I learned from many people who are more skilled than me, especially about structured thinking in mechanical design. I learned a lot about mechanical design optimization, surfacing, and calculating buoyancy and drag from design optimization.

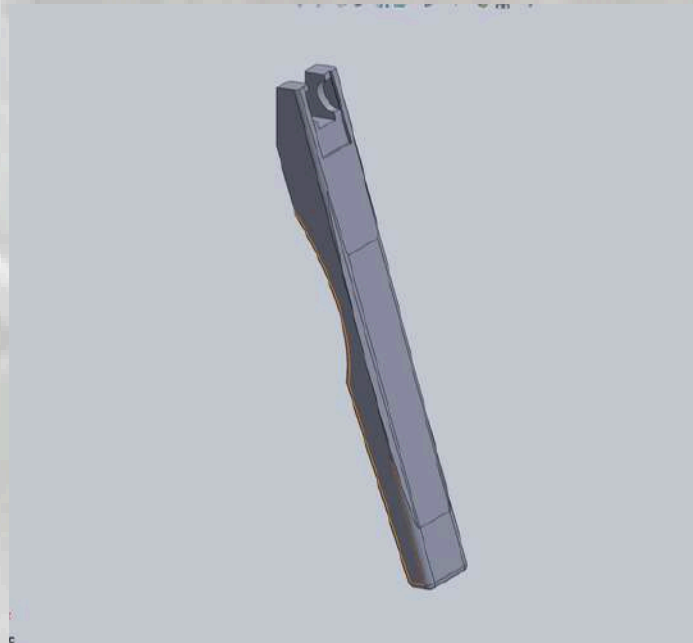
What Motivates me to do more?

My curiosity about something I didn't know before. I saw that all my friends were very skilled in their fields, so if I didn't hesitate to learn from them, I would become better in my field and be able to contribute more.

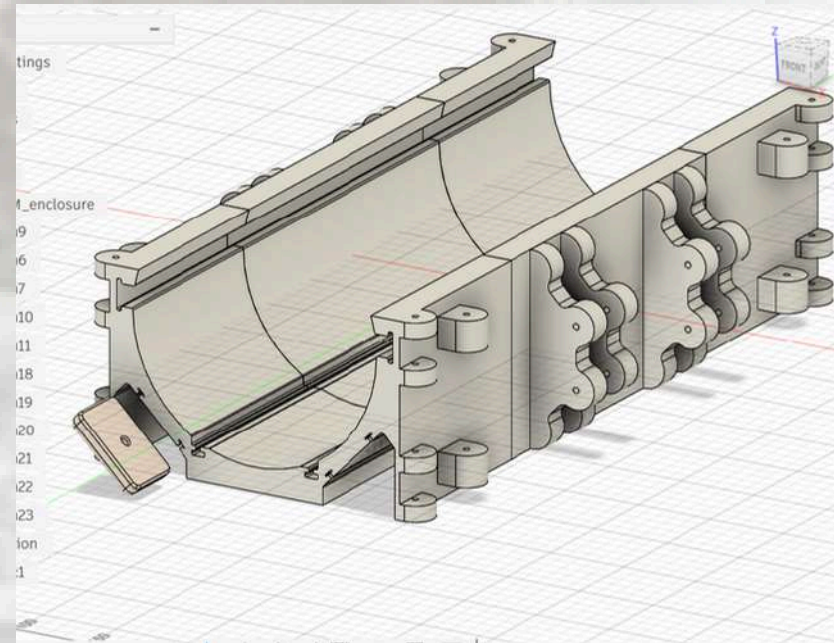
Current and Future Plan

The shell on the AUV II will be optimized, my plan is to modify the shell design to have good cable management and not affect drag performance. The robot will be neater and more professional.

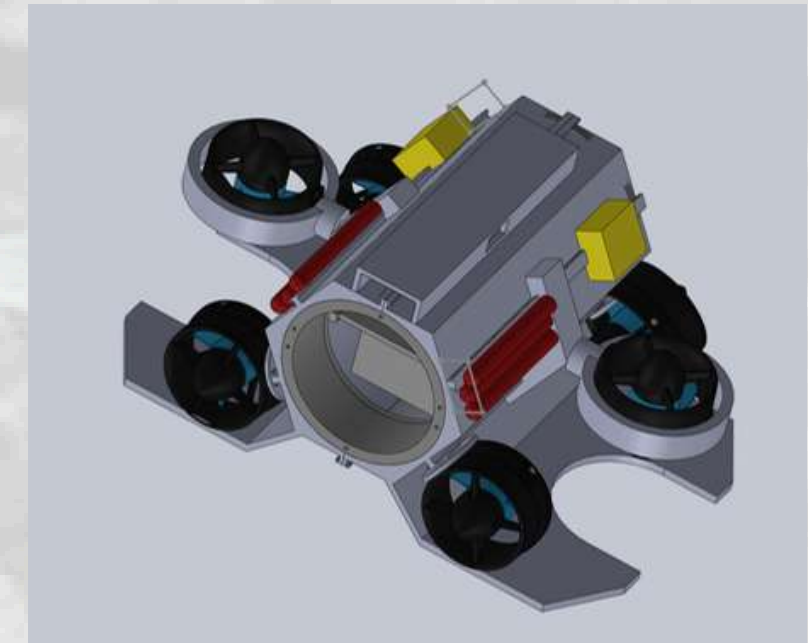
SOME OF MY CAD DESIGNS FOR sUNSWim!



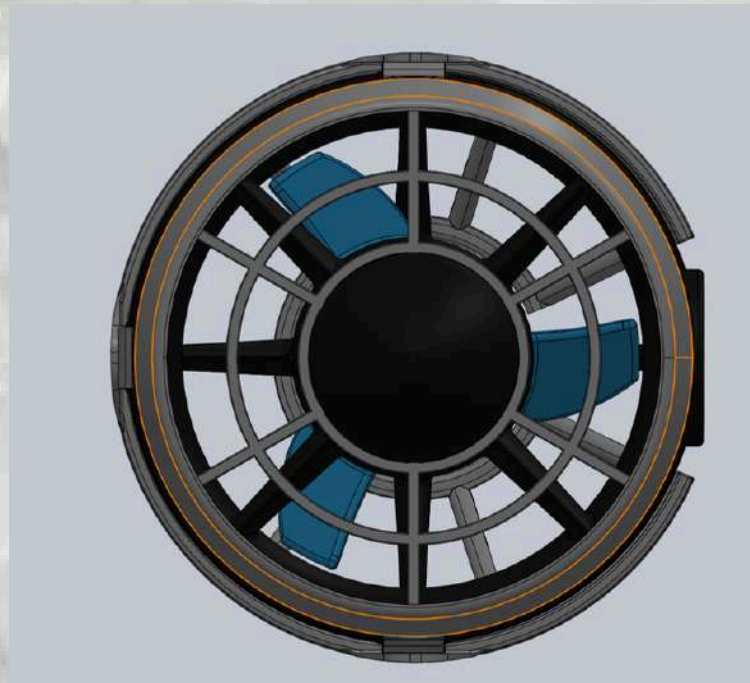
Vacuum Pump Handles



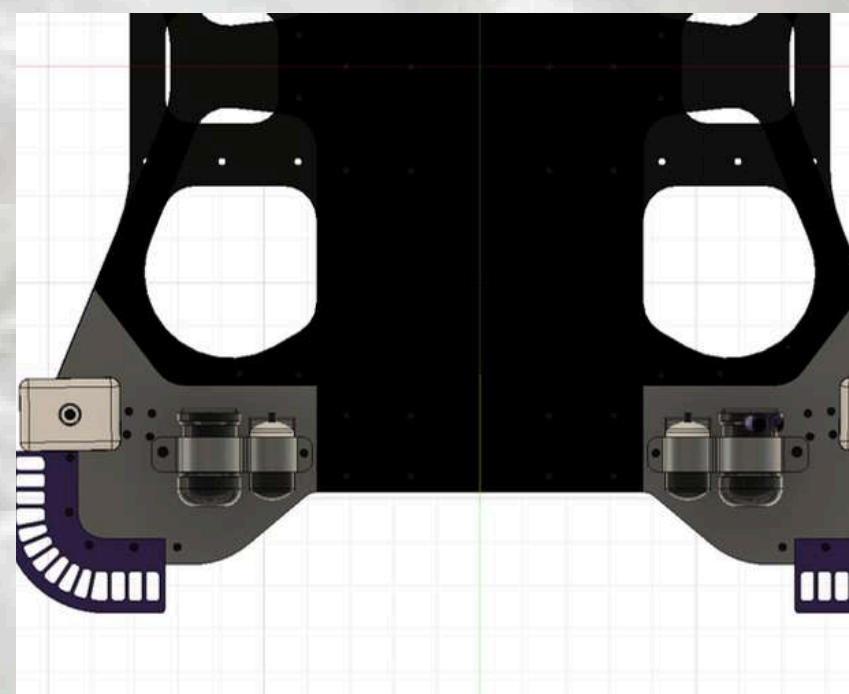
AUV III Update Chassis



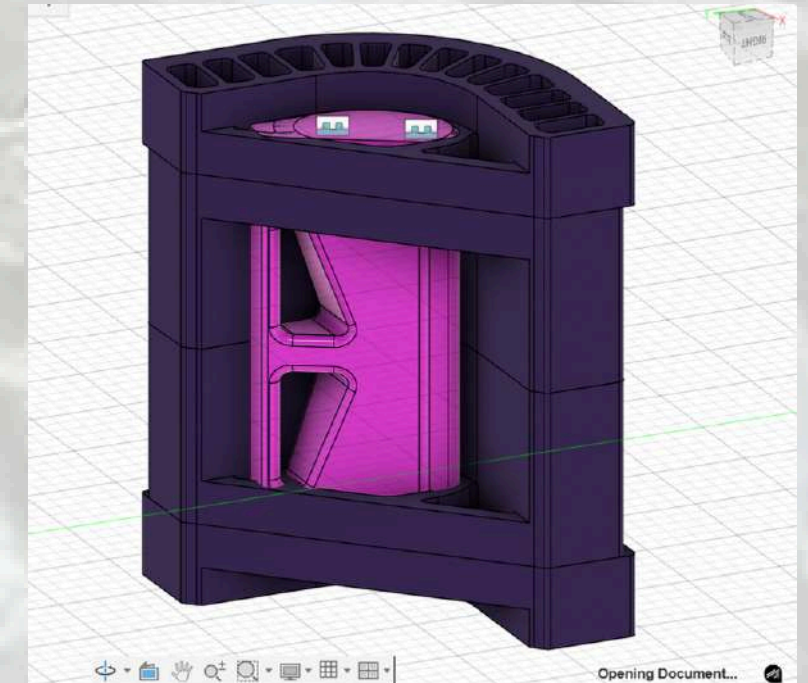
AUV III, Optimization of using 6 Propellers



Propeller Vortex Guards



Light and Hydrophone Mounting

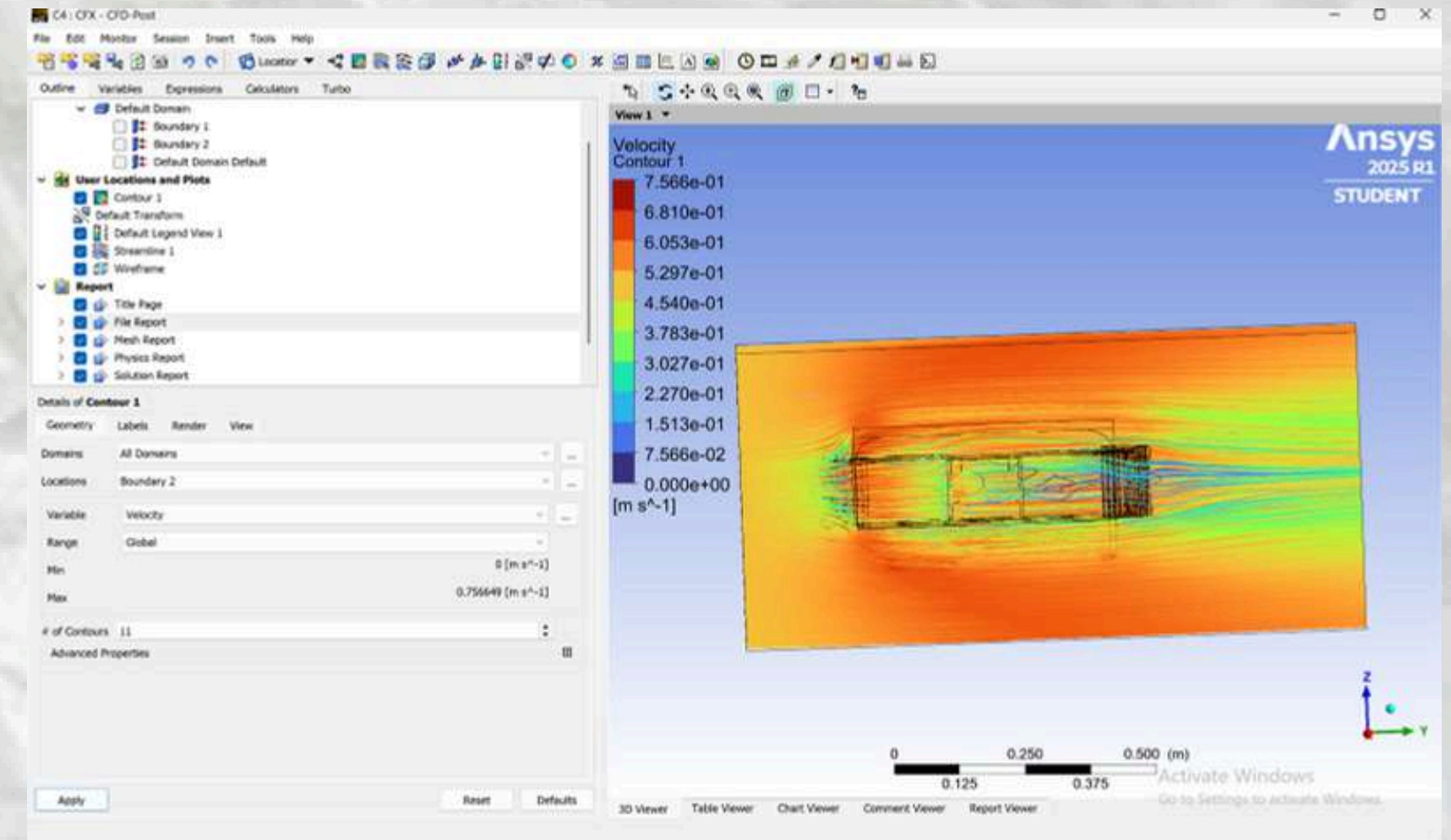
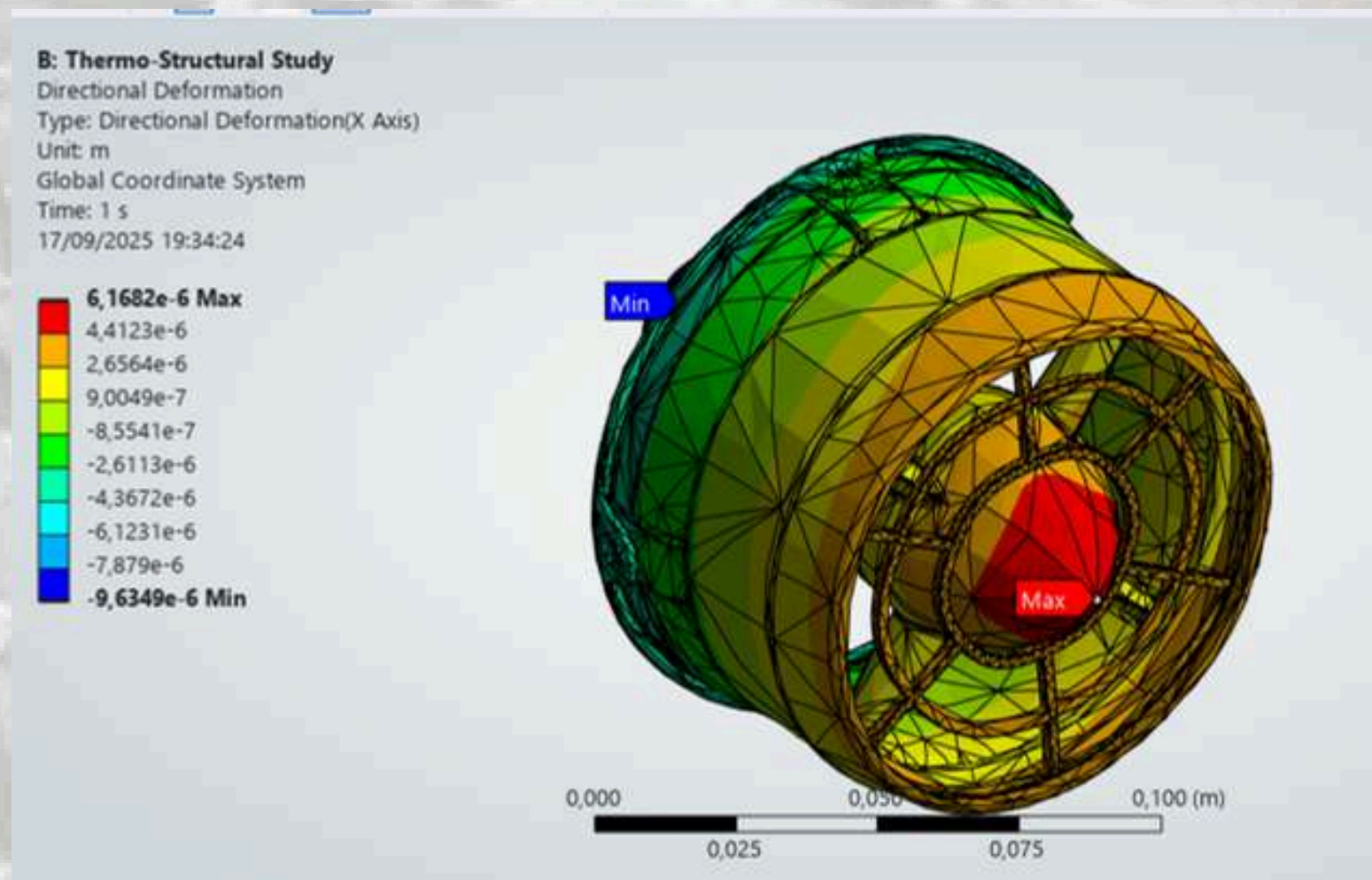


AUV II Bumper

CFD,FEA, & Thermal Validation

Some Design are need to be evaluated and simulated before real-life testing. This study (FEA and Thermal) validate how much the structure undertake the Hydrostatic Force up to 5 m with the Pool temperature. It isconcluded that the design can operate safely without fatigue.

Design Performance also evaluated to identify the Dynamics Performance of the ROV. This includes CFD study, considering the Buoyancy, Centre of Mass, Centre of Buoyancy, Drag Performance, and Turbulence Effect. We try to minimize the Drag coefficient and Turbulence. This study that I conduct on AUV II (Current Design) based on real Pool recorded condition, resulting in Drag Coeff about 0.91-0.97 in Air at 1 m/s.



Knee Brace for Osteoarthritis

Study Details

This study is part of the DESN2000 course, where our team conducted an analysis to validate the use of a knee brace for people with osteoarthritis.

Background

The main principle of this device is to assist the user's movement by requiring less effort and adding traction to reduce joint stress and strengthen the joint, thereby reducing the additional stress on the knee joint.

My Contribution

My efforts in this team were to validate the structural strength, specifically the gearing connecting the upper and lower knee braces, and to ensure their strength in withstanding the loads from traction and heavy loads imposed by the user.

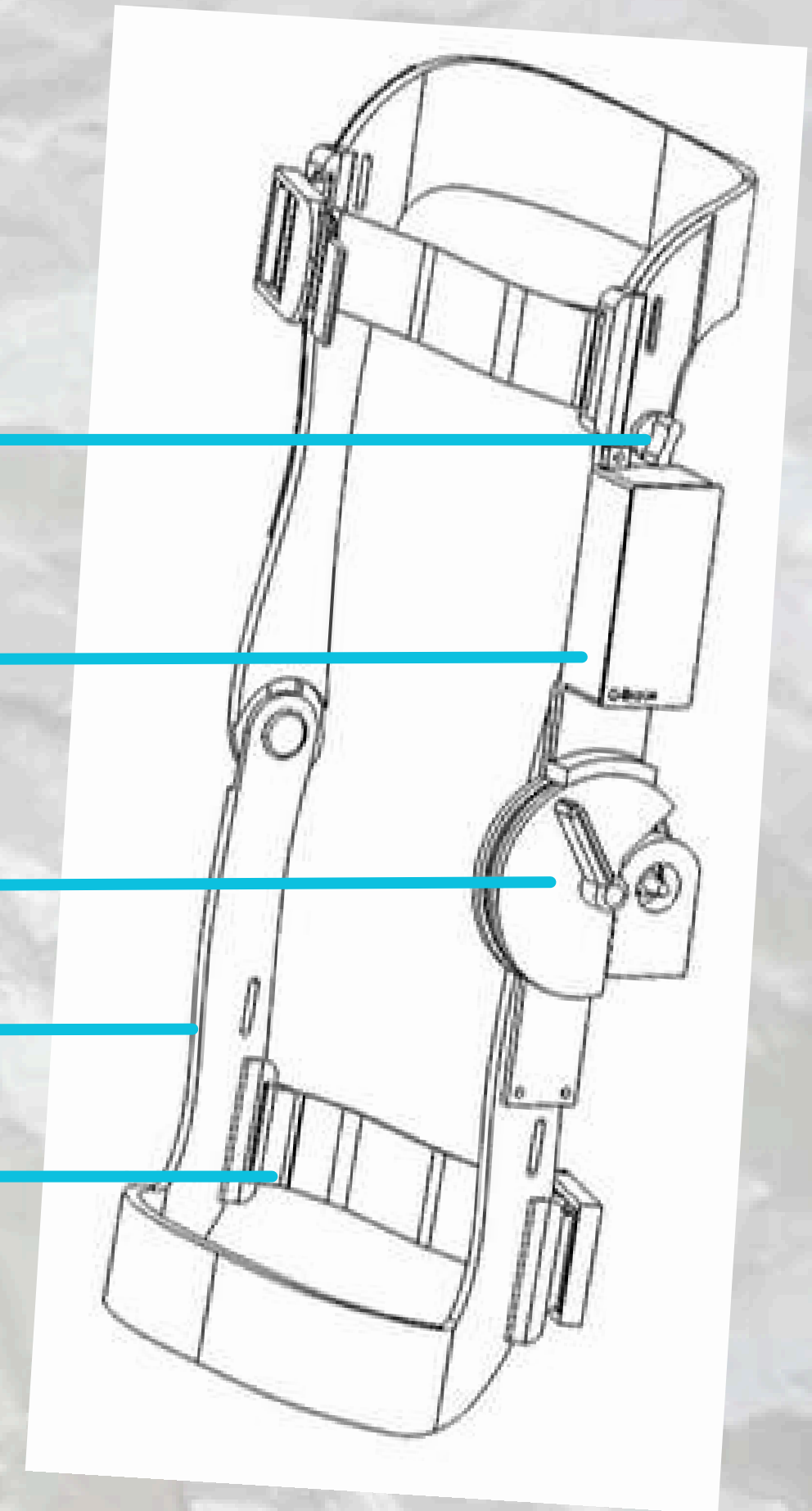
Traction System

Sensors

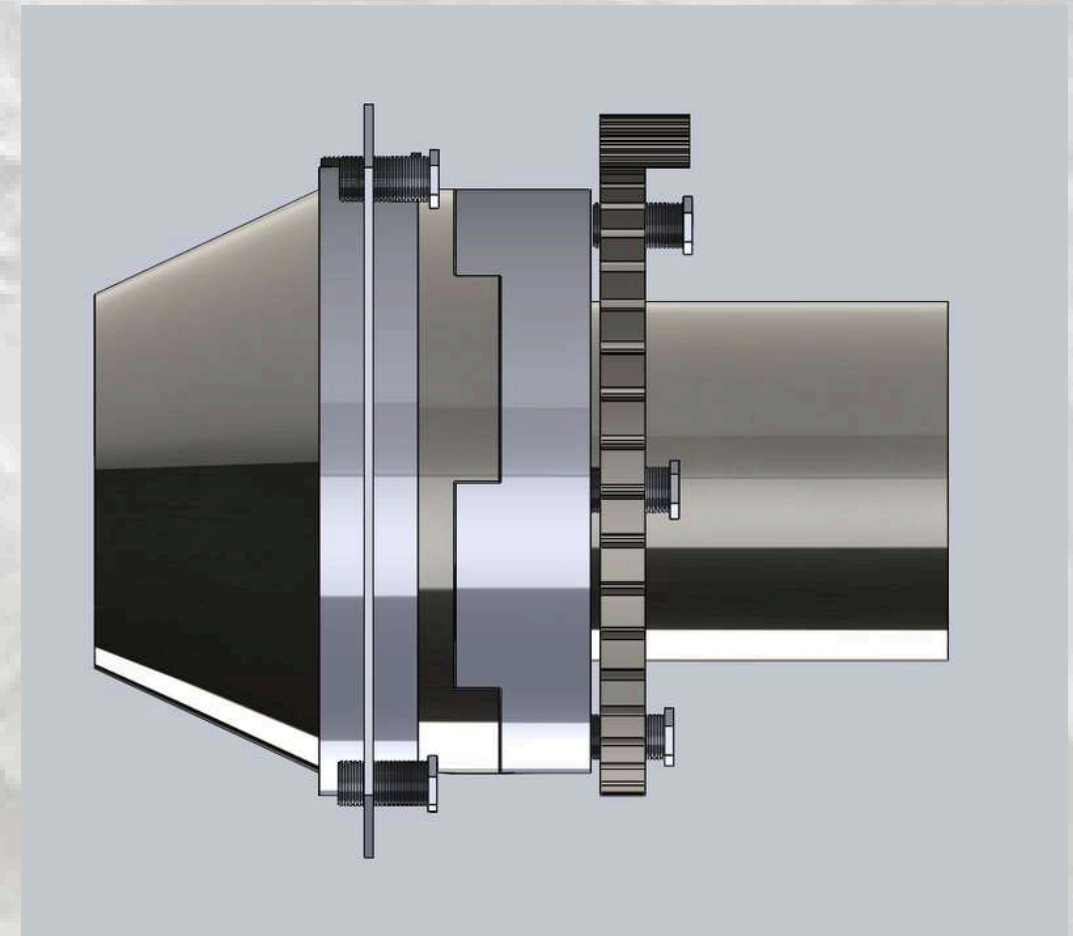
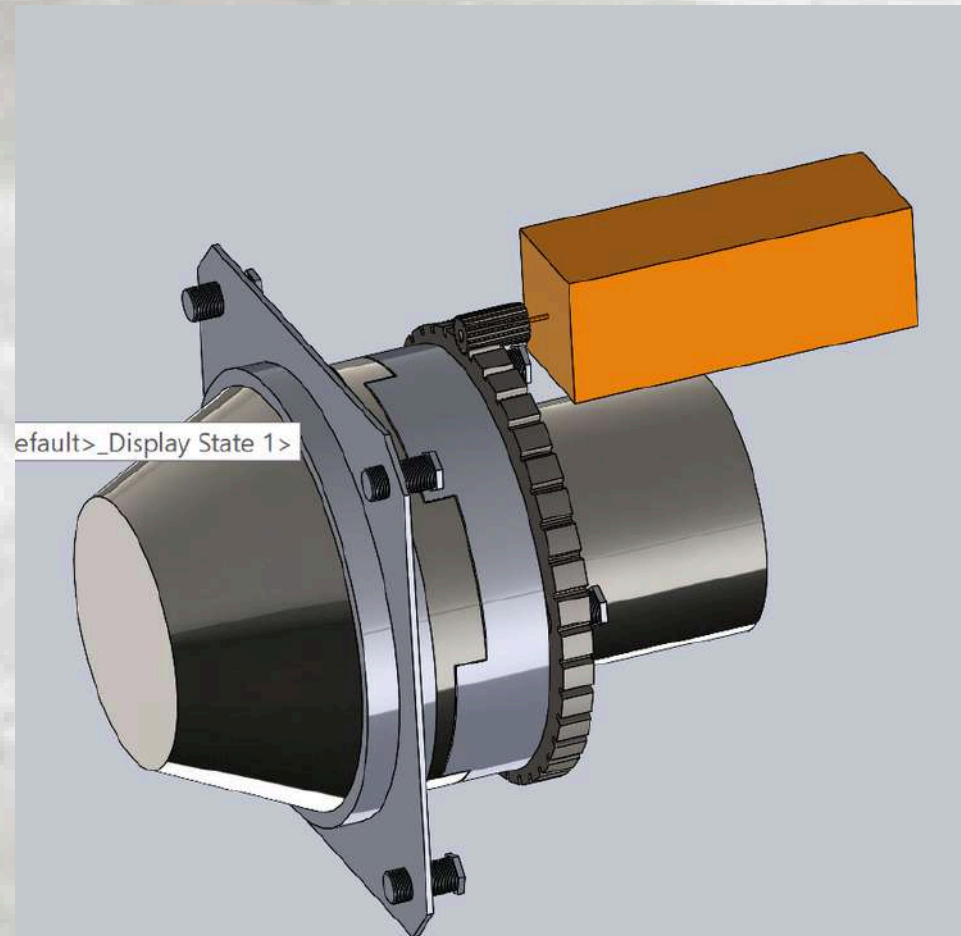
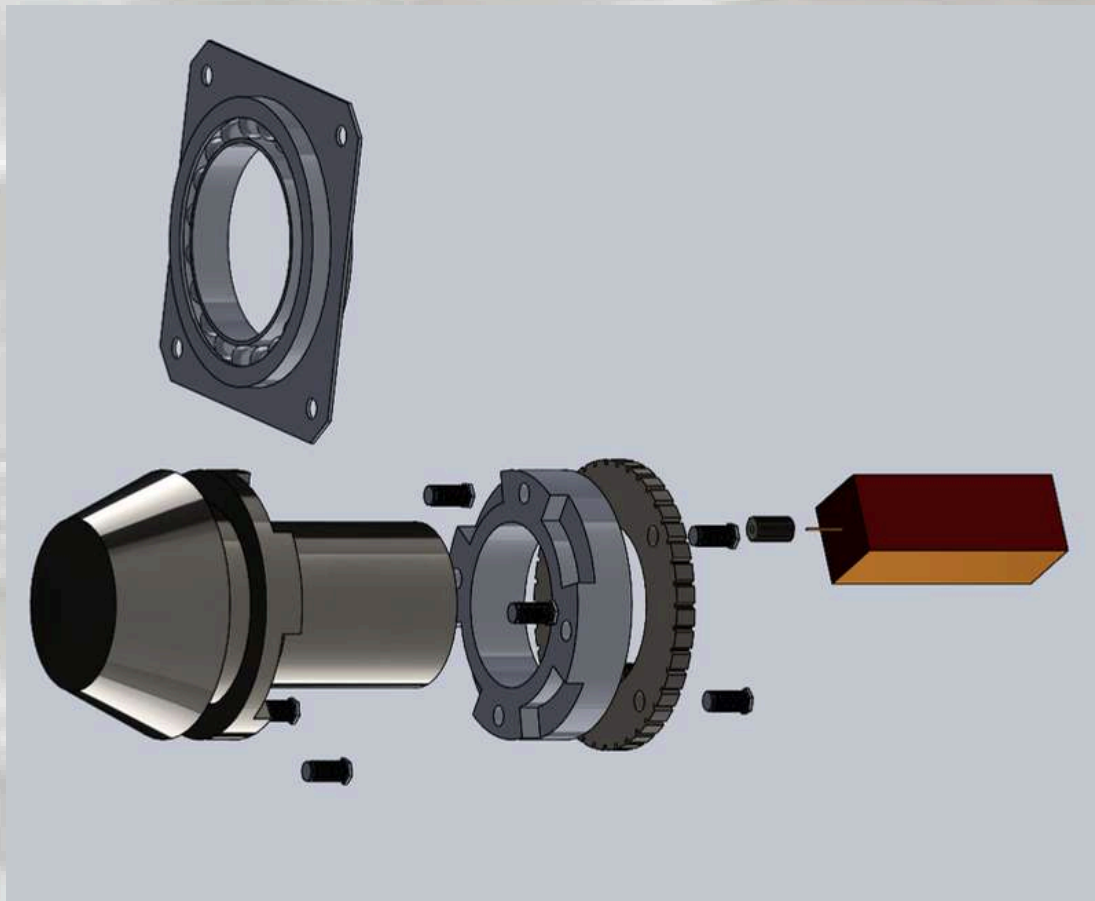
Hinge

Frame

Straps



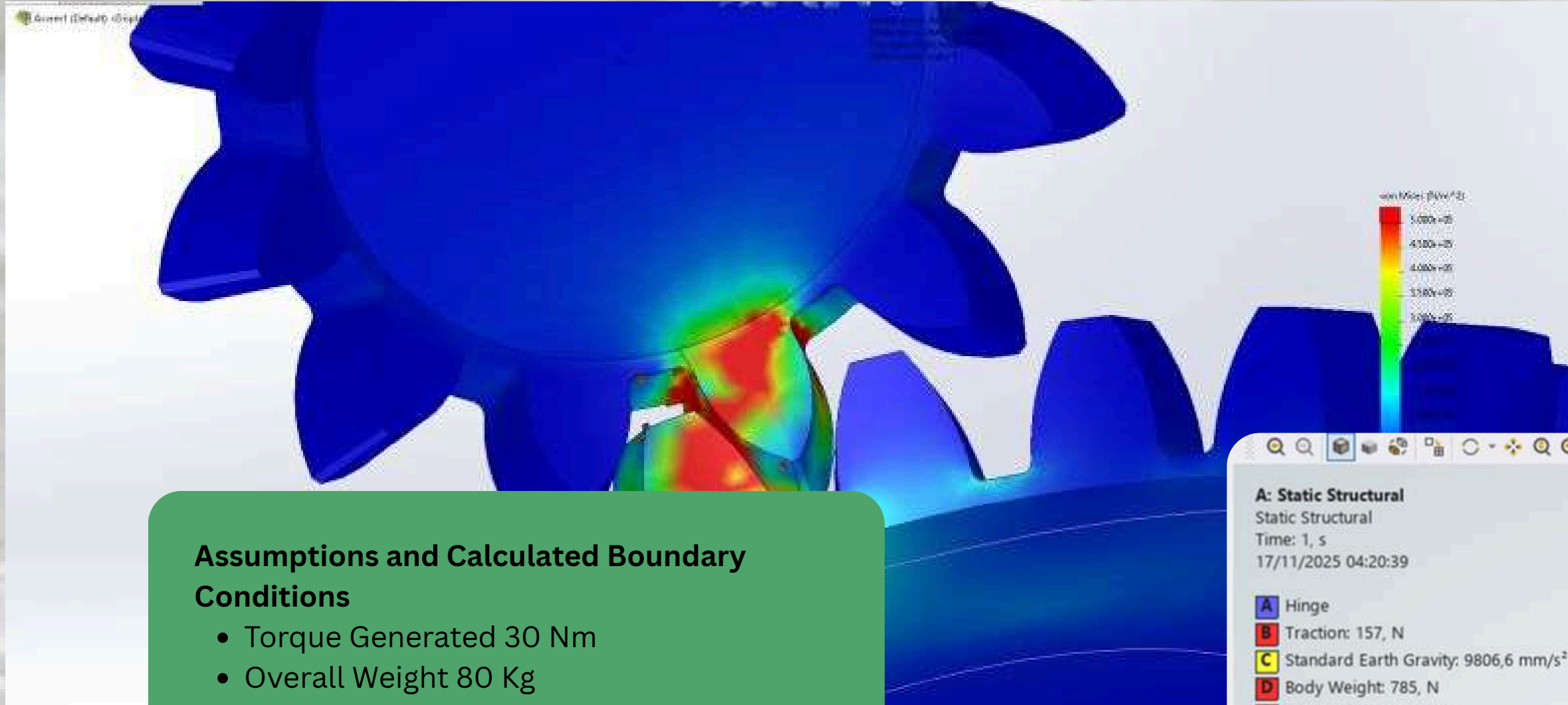
Hinge Traction Idea (Connecting Upper and Lower Knee Brace)



This subassembly is the principle of the gearing system I designed to connect the upper and lower sections of the brace. This gear section is the inner part of the hinge. It is expected that the gearing system will have strong resistance to overcome the torque exerted by the user during activities/extending the leg.

This design is also equipped with a Dog Clutch mechanism, enables us to change the component easily.

FEA on Gear Connection and Overall Frame

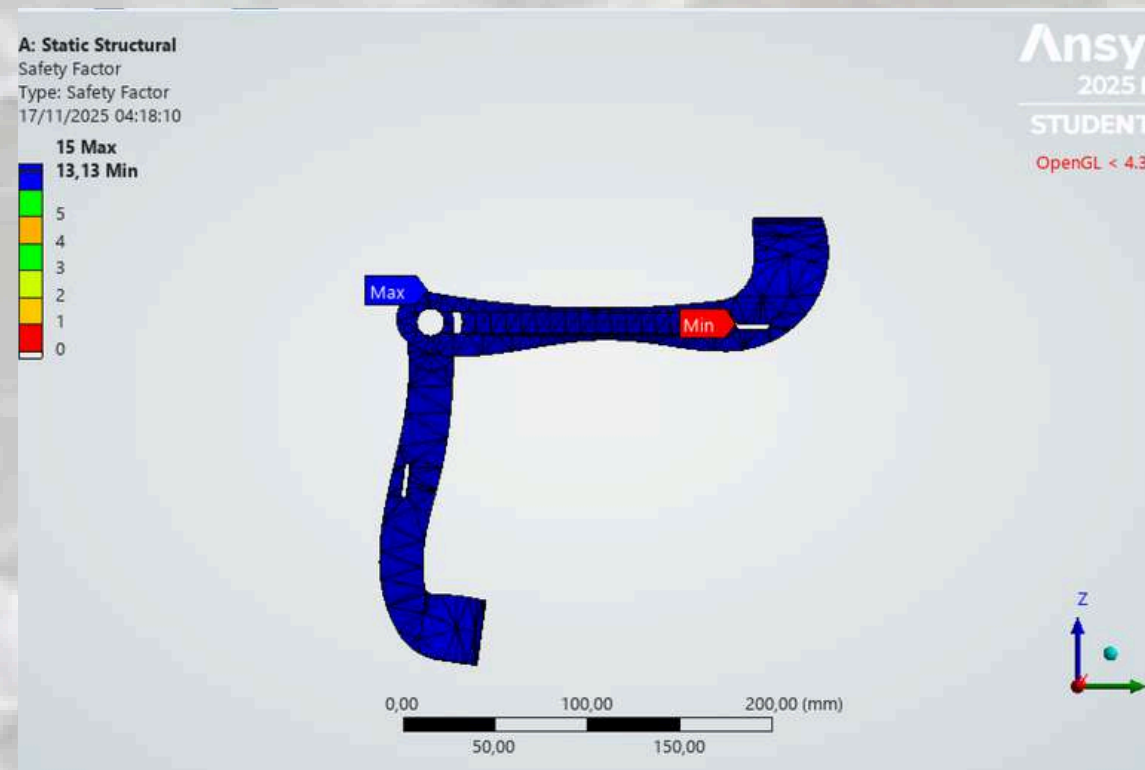


Assumptions and Calculated Boundary Conditions

- Torque Generated 30 Nm
- Overall Weight 80 Kg
- Traction (20% of body weight)
- No Compressive load
- Hinge was fixed to observe deformation
- Upper Brace was fixed Horizontally to simulate (sitting - standing)

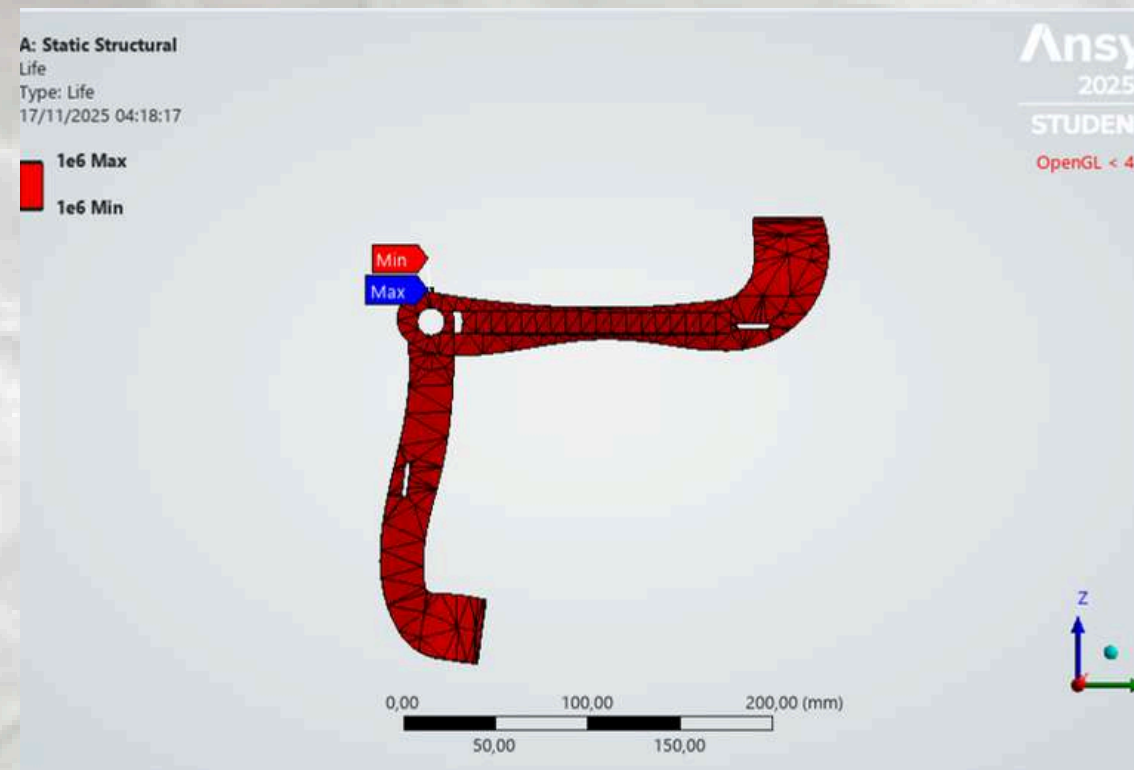


FEA Structural Analysis on Knee Brace



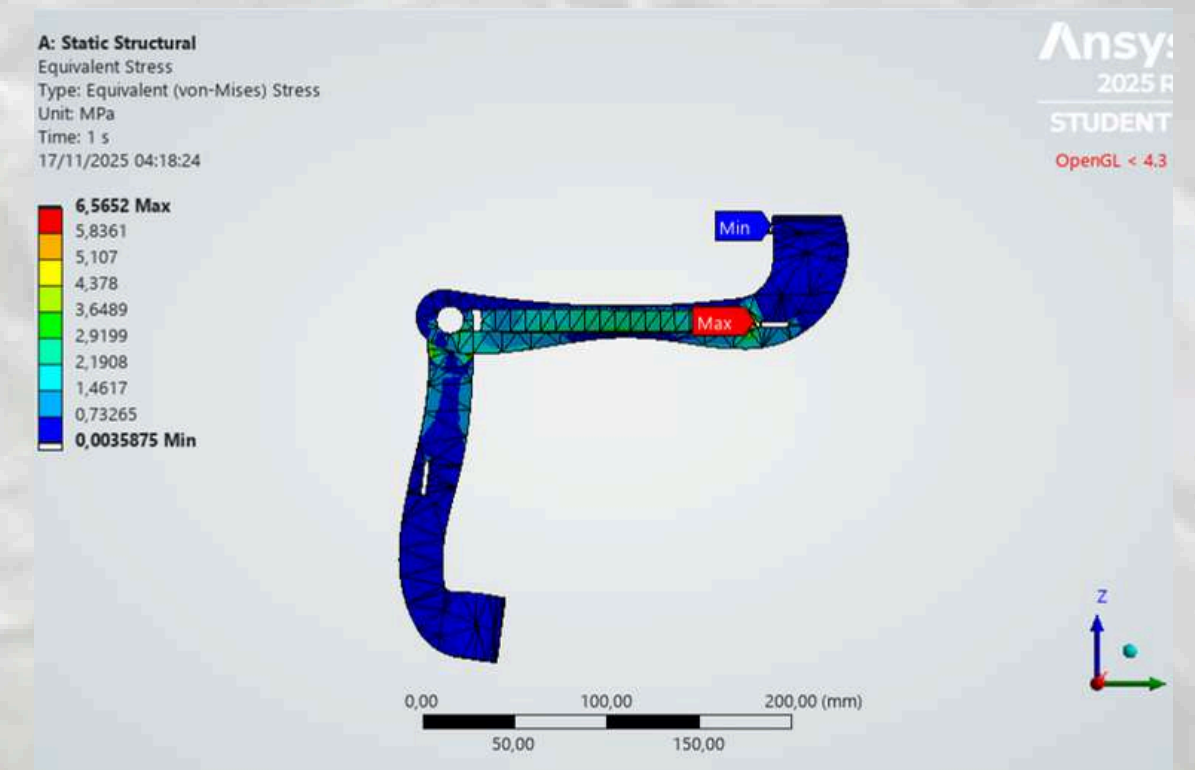
Factor of Safety > 13

- No risk of yielding or permanent deformation.
- High margin ensures reliability during repeated daily use.



Life Cycle > 1 Million

- Part operates in the infinite-life region of Aluminium 6061.
- Withstands over 1,000,000 loading cycles without fatigue failure.
- Suitable for long-term walking use (thousands of steps daily).



Max Stress : 6.56 MPa

- Lower than Aluminium 6061 Break Stress (UTS)
- far below its fatigue limit (95 MPa)
- Stress concentrations are small and localized (Near Hinge)